

Claudio Camolese

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🌐 Personal Website

in Personal Page

🐙 Github

Deep learning enthusiast.

Experience

- 2025  **AI Research Student UROP – Politecnico di Torino:**
Research project focused on optimizing inference of a Large Language Model (LLM) reasoning on edge devices.
Skills: *LLM, GenAI, Hugging Face*
- 2024 – 2025  **AI and Perception engineer – Squadra Corse/Driverless:**
Worked on the state estimation node using LiDAR odometry. Written documentation for a reinforcement learning agent integrated in Matlab and Simulink. Competed in the Formula SAE competitions.
Skills: *YOLO, LiDAR, ROS2, RViz, C++, Gazebo*

Education

- 2026  **Master thesis University of Cambridge.**
Investigating PINNs for reducing sim2real gap in deep reinforcement learning.
- 2024 – 2026  **Double M.Sc. Degree, specialization in AI.**
Double degree program at Politecnico di Torino, Italy and École nationale supérieure d'informatique et de mathématiques appliquées, France.
Relevant courses: *Robotics, Computer Vision, Natural Language Processing, GPU computing, Generative multimodal AI, Machine learning lab, Deep learning, Mathematics in machine learning, Robot learning, Numerical optimization.*
- 2021 – 2024  **BSc. Physics engineering.**
Bachelor's degree combining quantum physics, electronics and advanced mathematics, Relevant courses: *Quantum physics, Applied electronics, Electronic devices, Electromagnetic fields, Solid state physics.*
Thesis title: *Simulation of a neutron star using TOV equations.*

Projects

- 2025  **6D Pose Estimation.**
Development of a deep learning pipeline for estimating the 6D pose of objects from single RGB-D images. Used CNN and GraphNN.
-  **CNN in CUDA from scratch.**
Implement a full CUDA CNN training with only standard libraries.
-  **Generative AI for image generation.**
Implement diffusion, flow matching, GAN and VAE models for image generation.
-  **VLM for image description.**
Implement VLM to map an input image to a natural language description.

Projects (continued)

- **Physics Informed Neural Network for Heat diffusion.**
Implement PINN to solve thermal diffusion PDE equation.
- **Semantic segmentation on drone dataset.**
Implement a Deeplabv3+ architecture for image segmentation.
- **Binary semantic segmentation.**
Implement an end-to-end image segmentation pipeline using a Hourglass and a UNet.
- **Deep learning for 3D shape recovery.**
Reconstructs 3D shapes from multiple 2D silhouettes using voxel carving and an implicit 3D MLP.
- **LSTM for music generation.**
Use of an LSTM network for music generation and for merging music genres.
- **Deep Learning Optimization.**
Implementation and analysis of numerical optimization algorithms applied to large-scale deep learning problems. Focus on convergence analysis and performance.
- **Augmented Reality.**
AR application that detects a chessboard, calibrates the camera, estimates the pose and projects a 3D object onto the scene in real time.
- **Deep learning for sentiment classification.**
Train and evaluate a DistilBERT-based model for sentiment classification.
- **Deep Reinforcement learning Flappy Bird.**
Developed an AI Agent to play Flappy Bird game in gym environment.
- **Unsupervised learning for geometric shapes.**
Implemented the Spectral Clustering algorithm using the normalized Laplacian approach, based on the eigen decomposition of graph-related matrices derived from a similarity measure.

Skills

- Coding ■ Python, C++, CUDA, Matlab, Java
- Tools ■ LiDAR, YOLO, OpenCV, Git, Pytorch, ROS2, Gazebo, RViz, W&B, CometML, CMake, Docker, Linux, Spark, Hadoop, MongoDB, MLlib, Graphx, SQL, NoSQL, LabView, Arduino

Awards and Achievements

- **UROP undergraduate researcher open badge**
- **Scholarships:** Awarded with five Manager Italia Scholarships and one Stellantis N.V. scholarships for academic excellence.
- **GenNext entrepreneur:** Selected in the entrepreneurship program for top students
- **OpenCV certificate, VLM certificate, GNU/Linux program open badge**

Languages

- 2024 ■ **IELTS C1 English certificate.**
- 2017 ■ **DELFI B1 French certificate.**